

INTERVIEW SUBMISSION · INDEPENDENT PROTOTYPE

VC Retail Analytics

Synthetic Visual Comfort-flavored retail analytics: margin depth, seasonality, field grades, materials planning, and vendor scorecards – plus a governed assistant over warehouse metrics.

Open live demo →

<https://vc.datasharkbi.com> [GitHub →](#)

Prefer login **operator** / **vc-demo** · Access email OTP · then hard-refresh

SYNTHETIC DATA

NOT PRODUCTION

ALWAYS-ON

GOVERNED AI

Paul Rendleman

| Open the live demo (clickable)

DEMO · ACCESS-GATED · ALWAYS-ON (FLY)

<https://vc.datasharkbi.com>

Open demo →

- 01

Access OTP

Email code from Cloudflare Access (allowlisted guests).
- 02

Hard-refresh

Cmd+Shift+R so Analytics charts load (not cached TSV).
- 03

Sign in

/ for full Field & Supply.
- 04

Walk

Overview → Analytics (5 tabs) → Assistant → Catalog.

LOGIN	PASSWORD	USE WHEN
operator	vc-demo	Interview walkthrough – all territories, reps, vendors
dlr-0001	vc-demo	Prove dealer tenancy / session lock

Source: github.com/prendleman/vc-retail-analytics · Not Visual Comfort production systems.

THE OPPORTUNITY

Designer lighting retail, grounded in data

Public support handles FAQ and escalation – but dealer margin leakage, seasonal buy-ahead, rep quality, and vendor Prefer/Watch/Exit are missing from chat. This prototype closes that gap with governed warehouse metrics.

Ceiling

Wall

Lamps

Outdoor

Fans

Alabaster

Cordless

Channels

Consumer · Trade ·
Contract

Tenancy

Dealer scope from
session – never the
prompt

Honesty

Every panel is a
named metric query



Six analytics themes in the live app

CORE

Channel / family / region
bars + stock risk.

MARGIN

Margin %, price realization,
mix waterfall, low-margin
SKUs.

SEASON

Heatmap, YoY, lead-vs-
peak buy-ahead flags.

FIELD

Territories, coverage,
whitespace, rep A-D
grades.

SUPPLY

Days of cover, plan-vs-
season, vendor
Prefer/Watch/Exit.

Talk track

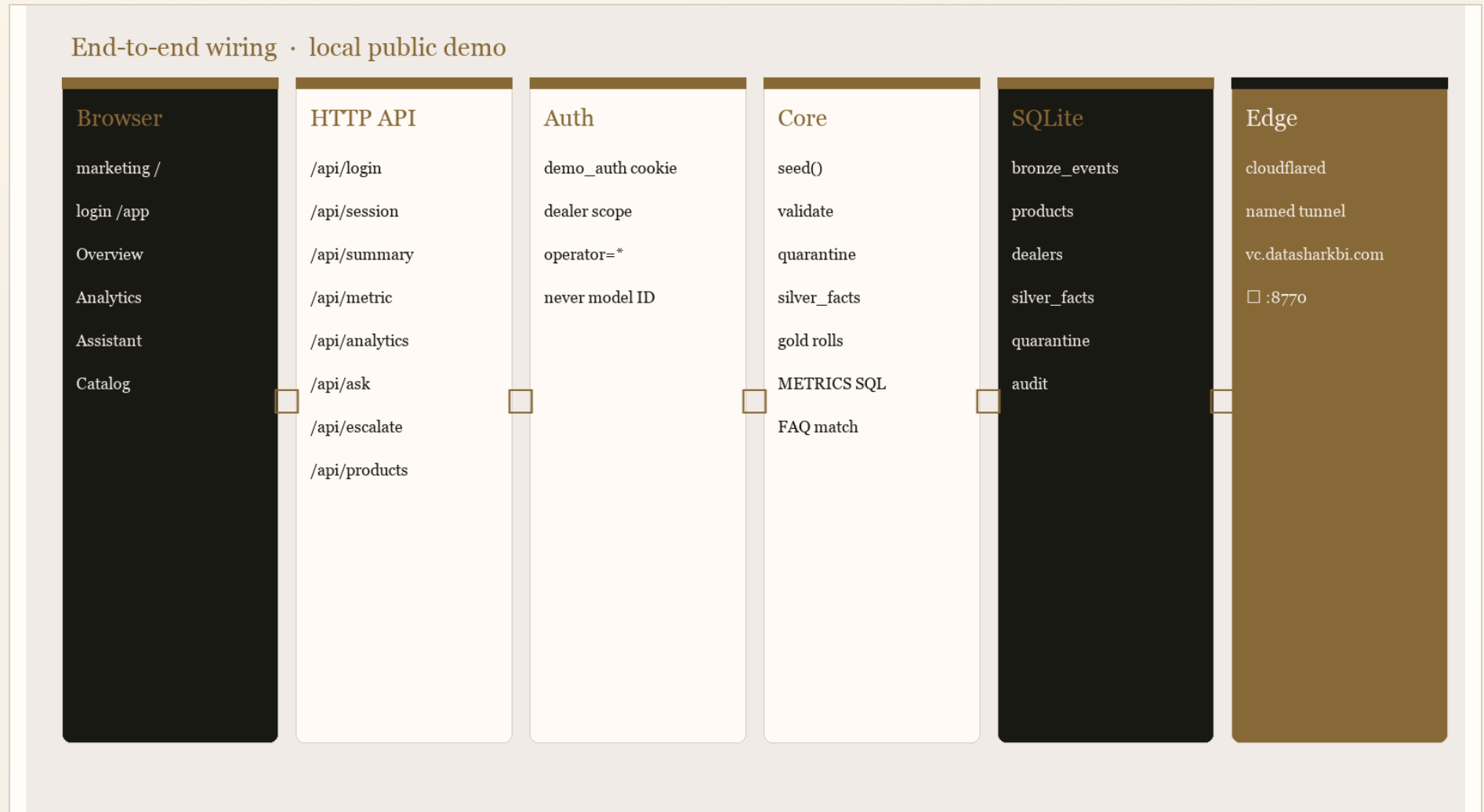
"Not vanity volume – margin discipline, seasonal buy-ahead before overseas
lead, Prefer / Watch / Exit on suppliers."

Insight callouts

Each Analytics section surfaces 2-4 governed findings (thinnest channel, mix
drag, Exit vendors, Pull-forward families) above charts and tables.

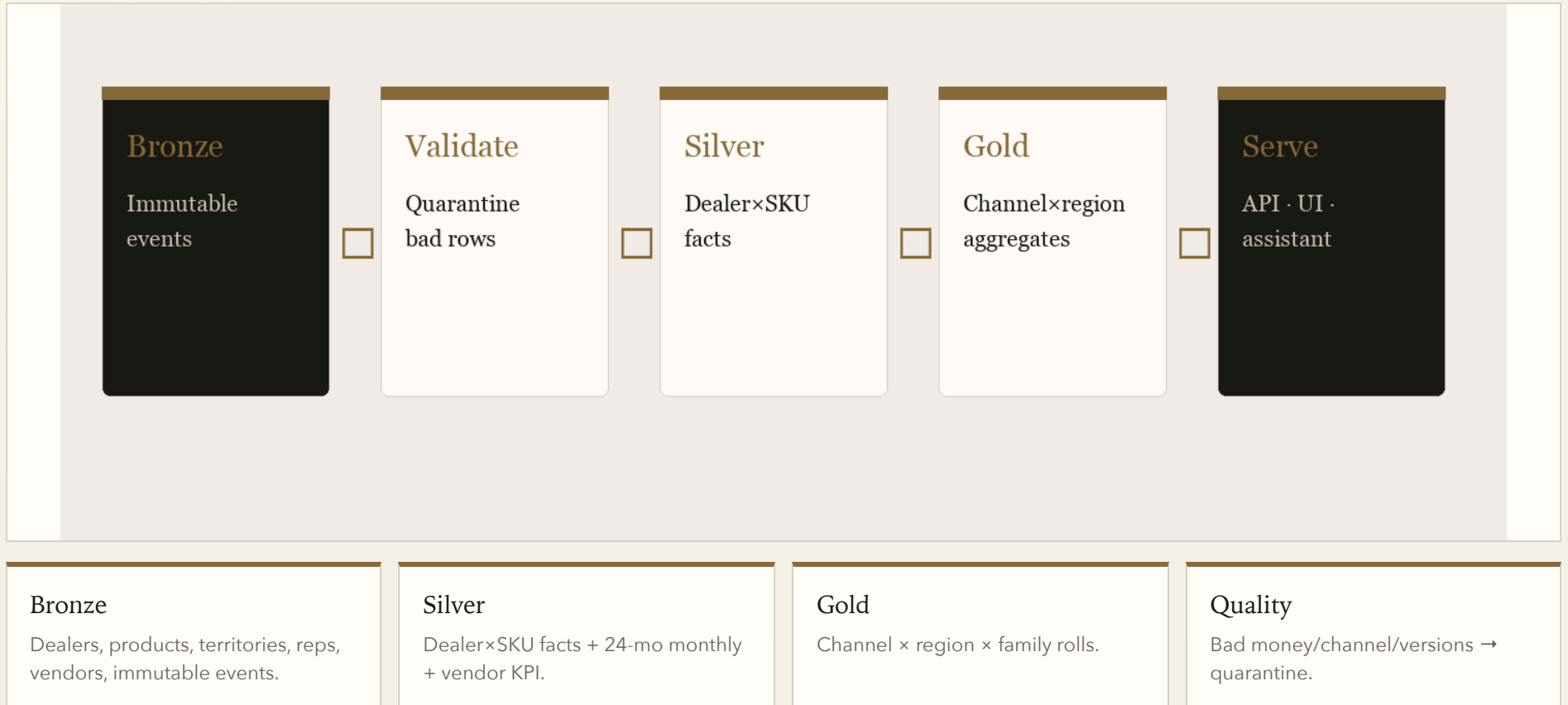
Assistant prompts: *show margin percent · show vendor scorecard · show rep grades · show lead vs peak*

Browser → Access → Fly → app + tunnel

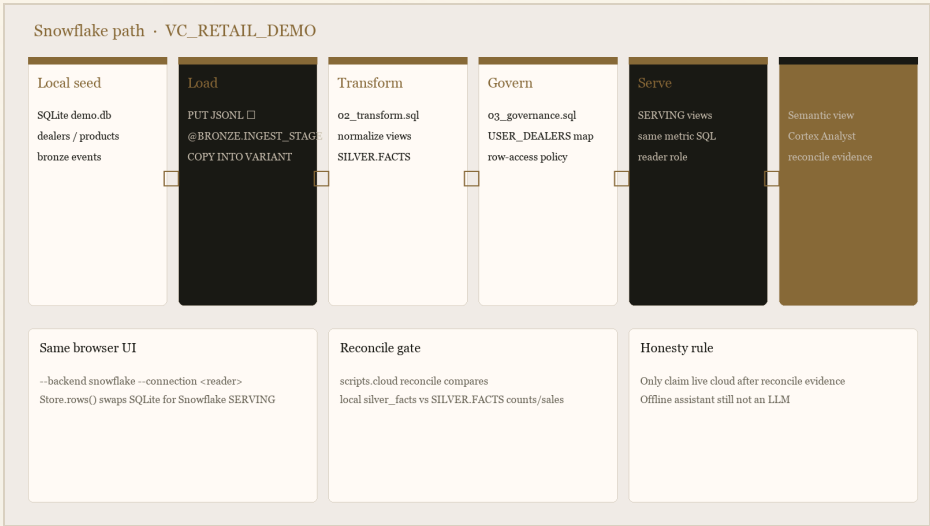


Always-on: Fly machine runs the stdlib app + `cloudflared` to **vc.datasharkbi.com** (laptop not required).

Bronze → Silver → Gold (+ monthly & vendors)



Snowflake — same product contracts



BRONZE	DEALERS, PRODUCTS, EVENTS, MONTHLY, VENDOR_KPI
SILVER	FACTS + QUARANTINE + MONTHLY
GOVERNANCE	USER_DEALERS + DEALER_SCOPE RAP
SERVING	Secure views + semantic view

Only claim “live Snowflake” after `scripts.cloud reconcile` evidence. Default demo is local SQLite on Fly.

One UI • two backends • closed metrics

Local / Fly

- `python3 -m app.server`
- SQLite `data/demo.db`
- Dealer WHERE from session cookie
- Hosted with Cloudflare Tunnel token

Snowflake

- `--backend snowflake`
- USE SCHEMA SERVING
- Same METRICS SQL shapes
- RAP enforces dealer scope in-warehouse

Governed metric contracts

portfolio silver_facts GROUP BY dealer	GET /api/metric
by_channel GROUP BY channel	GET /api/metric
by_family GROUP BY family	GET /api/metric
by_region GROUP BY region	GET /api/metric
stock_risk units_sold >= on_hand * 2	GET /api/metric
quality quarantine GROUP BY reason	GET /api/metric

API map & ask router

POST /api/login	Session cookie · dealer bind
GET /api/summary	Overview KPIs
GET /api/metric	Closed METRICS vocabulary
GET /api/analytics	Margin / season / field / supply bundle
POST /api/ask	today proposed · insight + table
POST /api/escalate	Structured handoff JSON
GET /api/products	Catalog attribute search

/api/ask routing · Today vs Proposed

TODAY mode

- match_faq(q)
- hit ☐ static FAQ + source id
- miss ☐ escalate stub
- form/phone pattern
- no warehouse tools

Escalation

- POST /api/escalate
- structured JSON packet
- lane hint + question
- demo handoff only

Shared guarantees

- Dealer filter from session cookie — never from the question text
- SYNTHETIC label · offline assistant is not an LLM

PROPOSED mode

- FAQ if policy-only
- else interpret_metric(q)
- metric_sql + dealer where
- review + declassification
- tool trace with SQL

Refuse / clarify

- unknown metric
- forecast invent
- closed vocabulary
- no hallucinated stock

Today vs Proposed • sample analytics

TODAY

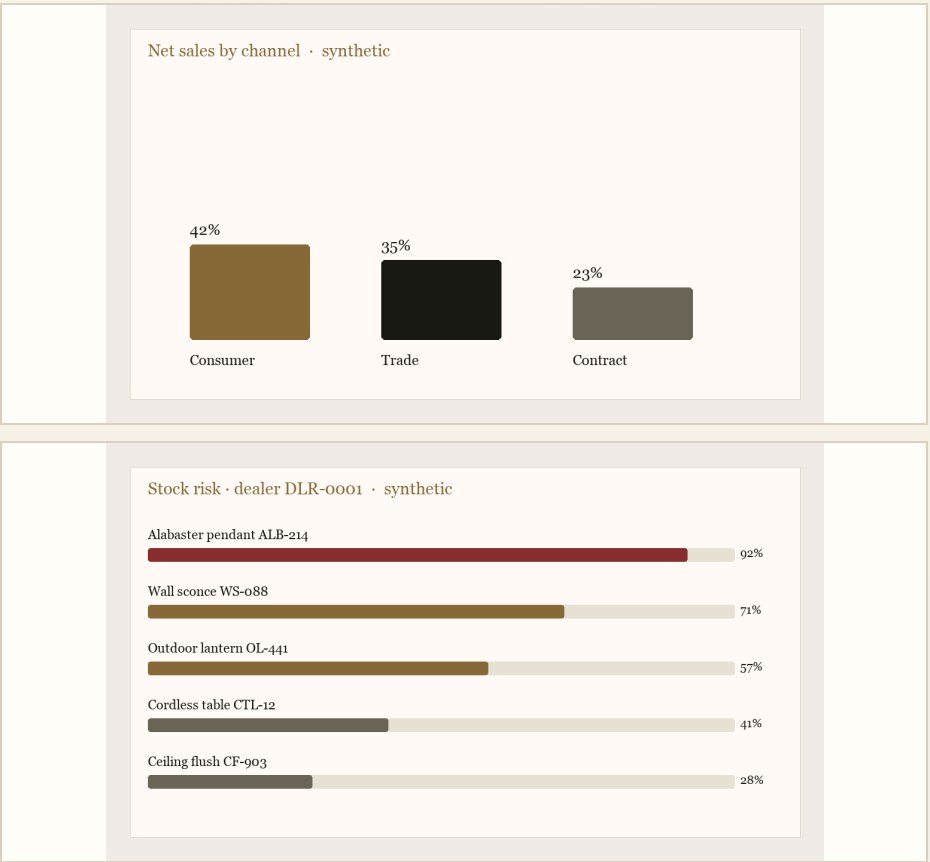
Site support pattern

- Form + phone lanes
- Static FAQ
- Human chat CS
- Escalate order status
- No warehouse metrics

PROPOSED

Governed retail AI

- Grounded FAQ + sources
- Catalog lookup
- Verified metric tools
- SQL in tool trace
- Structured escalate JSON



~10 minute walkthrough

<https://vc.datasharkbi.com>

Open live demo →

Today (site pattern)

- *custom order lead time* → FAQ snippet
- *what is open box* → policy
- *where is my order 12345* → escalate to form/phone
- Gap: no warehouse grounding

Proposed (governed)

- Login **operator** / vc-demo
- Analytics: Margin → Season → Field → Supply
- *show margin percent* → insight + chart + SQL
- *show vendor scorecard* → Prefer / Watch / Exit
- *show rep grades* → A-D composite
- *invent a forecast* → refuse

Offline assistant is a FAQ + metric router – **not an LLM**. Full script: `docs/DEMO_SCRIPT.md`.

| What this is — and isn't

Is

Independent synthetic portfolio piece · Visual
Comfort-flavored public brand language · always-
on governed demo

Isn't

Not Visual Comfort production data · not
commissioned client work · not a live LLM · not
Openflow

Submit pack

Live URL + Access · this PDF · GitHub source ·
operator walkthrough

[Demo](#) [Source](#)

INDEPENDENT · SYNTHETIC · GOVERNED

Ground retail AI in metrics.

Built so the team can see how dealer analytics and a trustworthy assistant should be wired – across margin, season, field, and supply.

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